Online Appendix

Not Only Extremes: How Moderate Rainfall Deficits Affect Child Health in Mexico

Zachary Serby

Table of Contents for Online Appendix

A Sample Sizes and Attrition **2**

B Continuous Dose **5**

C Heterogeneity by Sex and by Rural and Urban Residence **8**

A Sample Sizes and Attrition

Table OA1: Sample Sizes, Treated and Control Children

	Panel A: Binary DID, 2009 drought	
	Control	Treated
HAZ sample, children aged 15 or younger in 2005 and 2009		
Children, estimation sample	1,514	922
Children, Model 1 (basic DID)	1,512	920
Children, Model 3 (DID with controls)	1,077	678
Children, Model 5 (triple difference)	987	635
In 2002 remittance households	390	193
Households, estimation sample	944	578
2005 municipalities, estimation sample	87	46
WAZ sample, children aged 10 or younger in 2005 and 2009		
Children, estimation sample	696	388
Children, Model 1 (basic DID)	692	386
Children, Model 3 (DID with controls)	445	255
Children, Model 5 (triple difference)	400	234
In 2002 remittance households	168	90
Households, estimation sample	565	316
2005 municipalities, estimation sample	79	44
BMIZ sample, children aged 15 or younger in 2005 and 2009		
Children, estimation sample	1,487	896
Children, Model 1 (basic DID)	1,485	894
Children, Model 3 (DID with controls)	1,057	657
Children, Model 5 (triple difference)	968	619
In 2002 remittance households	381	189
Households, estimation sample	931	561
2005 municipalities, estimation sample	87	46
	Panel B: Continuous DID, any 2009 deficit	
	Control	Treated
HAZ sample, children aged 15 or younger in 2005 and 2009		
Children, continuous DID sample	829	1,607
In 2002 remittance households	236	438
In 2002 non-remittance households	437	738
Households	501	1,021
2005 municipalities	46	87
WAZ sample, children aged 10 or younger in 2005 and 2009		
Children, continuous DID sample	380	704
In 2002 remittance households	100	193
In 2002 non-remittance households	172	260
Households	304	577
2005 municipalities	42	81
BMIZ sample, children aged 15 or younger in 2005 and 2009		
Children, continuous DID sample	815	1,568
In 2002 remittance households	231	427
In 2002 non-remittance households	430	720
Households	496	996
2005 municipalities	46	87

	Panel C: Dose groups vs near normal	
	Control	Treated
Moderately dry ($-1.5 < z \leq -1$)		
Children, HAZ sample	1,493	488
Children, WAZ sample	683	217
Children, BMIZ sample	1,464	472
2005 municipalities, HAZ sample	89	23
2005 municipalities, WAZ sample	80	23
2005 municipalities, BMIZ sample	89	23
Severely dry or worse ($z \leq -1.5$)		
Children, HAZ sample	1,493	370
Children, WAZ sample	683	145
Children, BMIZ sample	1,464	364
2005 municipalities, HAZ sample	89	17
2005 municipalities, WAZ sample	80	16
2005 municipalities, BMIZ sample	89	17
Severely dry ($-2 < z \leq -1.5$)		
Children, HAZ sample	1,493	302
Children, WAZ sample	683	121
Children, BMIZ sample	1,464	296
2005 municipalities, HAZ sample	89	15
2005 municipalities, WAZ sample	80	14
2005 municipalities, BMIZ sample	89	15
Extremely dry ($z \leq -2$)		
Children, HAZ sample	1,493	68
Children, WAZ sample	683	24
Children, BMIZ sample	1,464	68
2005 municipalities, HAZ sample	89	2
2005 municipalities, WAZ sample	80	2
2005 municipalities, BMIZ sample	89	2

Notes: Counts are children, households or 2005 municipalities in each regression sample at the 2005 baseline. Estimation samples are children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009) who have a 2005 municipality. They are aged 15 or younger for HAZ and BMIZ and 10 or younger for WAZ in both survey waves. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. In Panel A, treated children's 2005 municipality had a drought in 2009, meaning May–October CHIRPS rainfall in the maize growing season was below its 1981–2004 20th percentile (Jessoe, Manning, and Taylor 2018; Shah and Steinberg 2017). Model 1 drops children alone in their interview-month cell. Continuous DID uses the full sample. Model 3 adds 2005–2008 droughts and 2002 household and 2005 child and mother controls. Model 5 adds 2002 remittances times post and drought times post times 2002 remittances. Model 5 children not in remittance households are the Model 5 count minus the remittance row. In Panel B, treated children's 2005 municipality had 2009 rainfall below its 1981–2004 mean (z below 0), where z is 2009 rainfall in standard deviations from that mean. Panel B gives sample sizes for the continuous DID. Its two remittance rows are children in 2002 remittance and non-remittance households, the two groups compared in Panel B of Table OA3. Rows do not add up to the total because children with missing 2002 remittances are in neither row. Panel C compares each dose group with near normal (z between -1 and 1) in the discrete-dose analysis. Dose levels use standardized rainfall cutoffs of the SPI adopted by the World Meteorological Organization (McKee, Doesken, and Kleist 1993; WMO 2012). Children in wet municipalities (z of 1 or above) are not in the discrete-dose analysis. Remittances are cash received in the last 12 months from parents, siblings or children living outside the household, from MxFLS-1 (2002).

Table OA2: Sample Attrition Across Waves

	Number of children (N)		
	MxFLS-1 (2002)	MxFLS-2 (2005)	MxFLS-3 (2009)
HAZ sample, children under 16			
DID (2005 and 2009)		3,582	2,436
Event studies (all three waves)	2,723	1,856	1,300
WAZ sample, children under 11			
DID (2005 and 2009)		1,895	1,084
Event studies (all three waves)	962	609	275
BMIZ sample, children under 16			
DID (2005 and 2009)		3,511	2,383
Event studies (all three waves)	2,569	1,719	1,203

Notes: Counts of children in each wave. The binary DID only uses MxFLS-2 (2005) and MxFLS-3 (2009). The continuous DID uses the same sample as the binary DID. The event studies use all three waves. Each row starts from the children who meet the regression sample rules in its first wave and could stay in the sample. They are under the age cutoff in every wave, 15 or younger for HAZ and BMIZ and 10 or younger for WAZ. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Later columns count the children still in the sample, so the drop across columns is attrition. This is not the same as the children the regressions drop as singletons, with only one wave in the sample, as these also include children who aged out or are only in a later wave. The DID 2009 count is the estimation sample, and the event study 2009 count is the children in all three waves. For children not measured in a later wave, I estimate their age in that wave to see whether they aged out. Their age comes from their birth date and their household's MxFLS-3 interview date, since MxFLS-3 interviews ran over several years. Children not in MxFLS-3 or whose birth date does not match their reported age get their last measured age plus the median gap between waves. This gap is 4 years from 2005 to 2009 and 8 years from 2002 to 2009. Among children measured in both waves, 92 and 86 percent of gaps are within a year of the median. A one-year error only matters for children near the age cutoff. This way, children who simply aged out are not counted as attrition.

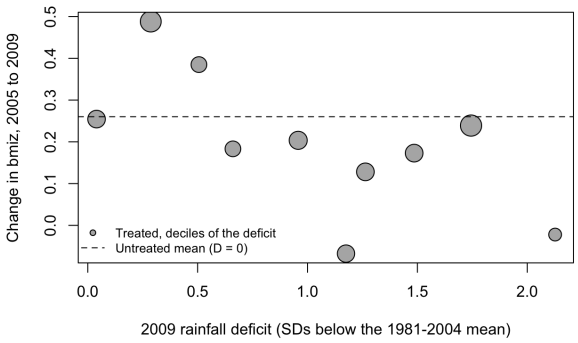
B Continuous Dose

Table OA3: Continuous DID and Remittances as a Moderator

	HAZ (1)	WAZ (2)	BMIZ (3)
Panel A: Continuous DID, any 2009 rainfall deficit			
Average ATT	-0.077	0.028	-0.045
	(0.101)	(0.097)	(0.069)
	[0.462]	[0.777]	[0.518]
Average ACRT	-0.007	-0.033	0.029
	(0.137)	(0.173)	(0.130)
Children	2,436	1,084	2,383
Panel B: Average ATT by 2002 remittances			
No 2002 remittances	-0.080	0.050	-0.042
	(0.107)	(0.152)	(0.091)
2002 remittances	-0.279	-0.030	0.031
	(0.182)	(0.190)	(0.116)
Difference, remittances minus none	-0.199	-0.081	0.074
	(0.179)	(0.218)	(0.145)
Children	1,849	725	1,808
Panel C: Dry vs near normal by 2002 remittances			
Dry ($z \leq -1$)	-0.054	-0.160	-0.251**
	(0.114)	(0.200)	(0.109)
	[0.657]	[0.543]	[0.027]
Dry × 2002 remittances	-0.135	-0.220	0.100
	(0.133)	(0.216)	(0.207)
	[0.317]	[0.329]	[0.653]
2002 remittances	-0.040	0.088	0.039
	(0.109)	(0.134)	(0.070)
	[0.722]	[0.524]	[0.577]
Children	1,773	693	1,734

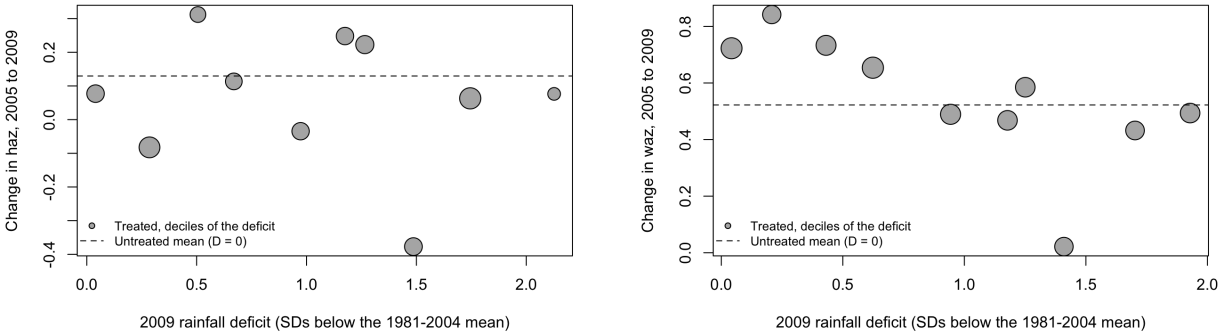
Notes: Coefficients are effects on the 2005 to 2009 change in the WHO z-score for the binary DID estimation samples. The dose is the 2009 rainfall deficit, max(0, -z), where z is 2009 May–October CHIRPS rainfall in the child's 2005 municipality in standard deviations from its 1981–2004 mean (Jessoe, Manning, and Taylor 2018). Panel A compares children with any 2009 deficit (z < 0) with children without one using the contdid package (Callaway, Goodman-Bacon, and Sant'Anna 2026). Panel B gives the average ATT for children in 2002 remittance and non-remittance households and their difference. Panel C interacts dry ($z \leq -1$) versus near-normal rainfall with 2002 remittances using the discrete-dose regression in equation 4.1 (Callaway, Goodman-Bacon, and Sant'Anna 2024). Remittances are cash received in the last 12 months from parents, siblings, or children outside the household in MxFLS-1 (2002). The average ATT requires parallel trends and the average ACRT requires strong parallel trends (Callaway, Goodman-Bacon, and Sant'Anna 2024). No regression includes controls or fixed effects. Standard errors clustered by 2005 municipality are in parentheses. For the average ACRT they come from a multiplier bootstrap of the contdid influence functions. Wild cluster bootstrap p-values are in brackets (MacKinnon, Nielsen, and Webb 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Figure OA1: Change in BMI-for-Age by 2009 Rainfall Deficit, Continuous Dose



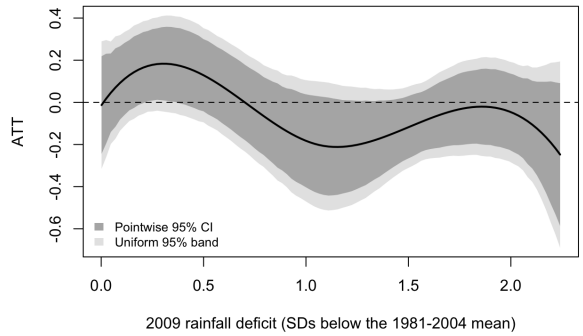
Notes: Mean 2005-to-2009 change in BMIZ by decile of the 2009 rainfall deficit $D = \max(0, -z)$ for treated children with $D > 0$. Circle size is proportional to the number of children in the decile. The dashed line is the mean change for untreated children ($D = 0$).

Figure OA2: Change in Height-for-Age and Weight-for-Age by 2009 Rainfall Deficit, Continuous Dose



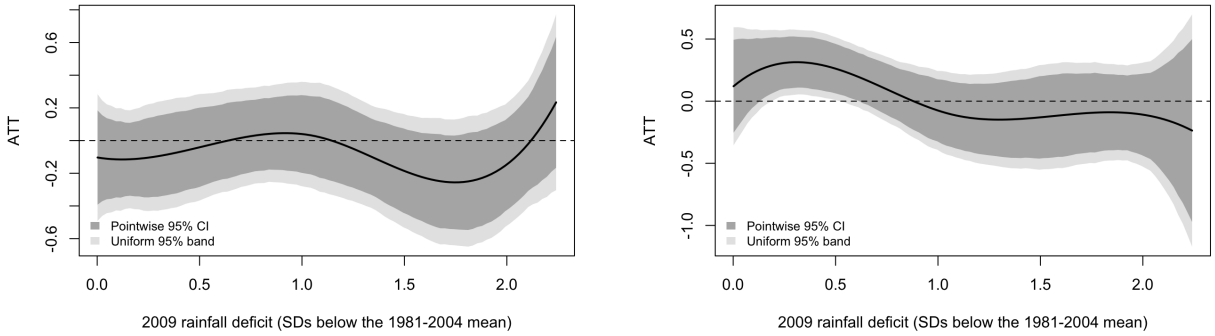
Notes: Left: HAZ. Right: WAZ. See Figure OA1 notes.

Figure OA3: ATT by 2009 Rainfall Deficit, BMI-for-Age, Continuous Dose



Notes: $ATT(d)$ is estimated with a cubic B-spline in the dose with one interior knot and evaluated over the range of treated doses using the contdid package (Callaway, Goodman-Bacon, and Sant'Anna 2026). The dark band is the pointwise 95 percent confidence interval and the light band is the uniform 95 percent band. Both come from a multiplier bootstrap of the influence functions clustered by 2005 municipality.

Figure OA4: ATT by 2009 Rainfall Deficit, Height-for-Age and Weight-for-Age, Continuous Dose



Notes: Left: HAZ. Right: WAZ. See Figure OA3 notes.

C Heterogeneity by Sex and by Rural and Urban Residence

Table OA4: 2009 Drought and Height-for-Age, Binary DID, Boys

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.123	-0.108	-0.108	0.228	-0.075
	(0.123)	(0.130)	(0.146)	(0.237)	(0.142)
Drought × post × 2002 remittances					-0.301
					(0.257)
2002 remittances × post					-0.254**
					(0.123)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,201	1,201	869	353	806

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for boys, using sex reported in the 2005 household roster, aged 15 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include child sex. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA5: 2009 Drought and Weight-for-Age, Binary DID, Boys

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.219	-0.175	-0.264	-0.330	-0.064
	(0.153)	(0.149)	(0.182)	(0.358)	(0.196)
Drought × post × 2002 remittances					-0.225
					(0.283)
2002 remittances × post					0.084
					(0.179)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	551	551	367	103	331

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for boys, using sex reported in the 2005 household roster, aged 10 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include child sex. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA6: 2009 Drought and BMI-for-Age, Binary DID, Boys

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.170*	-0.168	-0.269**	-0.262	-0.336*
	(0.099)	(0.119)	(0.131)	(0.180)	(0.175)
Drought × post × 2002 remittances					0.416
					(0.322)
2002 remittances × post					-0.057
					(0.111)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,170	1,170	847	346	787

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for boys, using sex reported in the 2005 household roster, aged 15 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include child sex. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA7: 2009 Drought and Height-for-Age, Binary DID, Girls

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.046	0.020	0.111	-0.055	0.035
	(0.101)	(0.116)	(0.131)	(0.202)	(0.139)
Drought × post × 2002 remittances					-0.060
					(0.176)
2002 remittances × post					-0.143
					(0.115)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,220	1,220	878	377	811

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for girls, using sex reported in the 2005 household roster, aged 15 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include child sex. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA8: 2009 Drought and Weight-for-Age, Binary DID, Girls

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.259**	-0.282**	-0.379**	-0.184	-0.265
	(0.111)	(0.119)	(0.167)	(0.288)	(0.197)
Drought × post × 2002 remittances					-0.401
					(0.302)
2002 remittances × post					0.097
					(0.189)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	520	520	324	115	295

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for girls, using sex reported in the 2005 household roster, aged 10 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include child sex. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA9: 2009 Drought and BMI-for-Age, Binary DID, Girls

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.112	-0.160	-0.313***	-0.200	-0.184
	(0.077)	(0.097)	(0.116)	(0.167)	(0.114)
Drought × post × 2002 remittances					-0.342*
					(0.200)
2002 remittances × post					0.065
					(0.098)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,199	1,199	861	364	795

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for girls, using sex reported in the 2005 household roster, aged 15 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include child sex. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA10: 2009 Drought and Height-for-Age, Binary DID, Rural Children

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.045	0.189	0.190	0.472	0.273*
	(0.162)	(0.173)	(0.163)	(0.289)	(0.159)
Drought × post × 2002 remittances					-0.278
					(0.288)
2002 remittances × post					-0.006
					(0.130)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,016	1,016	732	272	714

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for rural children (2005 household locality), aged 15 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include rural. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA11: 2009 Drought and Weight-for-Age, Binary DID, Rural Children

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.178	-0.074	-0.276	0.532	-0.131
	(0.180)	(0.172)	(0.264)	(0.334)	(0.281)
Drought × post × 2002 remittances					-0.362
					(0.332)
2002 remittances × post					0.216
					(0.225)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	414	414	265	85	260

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for rural children (2005 household locality), aged 10 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include rural. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA12: 2009 Drought and BMI-for-Age, Binary DID, Rural Children

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.096	-0.138	-0.208	-0.244	-0.197
	(0.112)	(0.127)	(0.180)	(0.254)	(0.204)
Drought × post × 2002 remittances					-0.035
					(0.255)
2002 remittances × post					0.086
					(0.095)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	998	998	719	264	701

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for rural children (2005 household locality), aged 15 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include rural. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA13: 2009 Drought and Height-for-Age, Binary DID, Urban Children

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.102	-0.136	-0.088	-0.021	-0.243*
	(0.106)	(0.110)	(0.125)	(0.200)	(0.122)
Drought × post × 2002 remittances					0.032
					(0.167)
2002 remittances × post					-0.331***
					(0.121)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,412	1,412	1,020	455	904

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for urban children (2005 household locality), aged 15 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include rural. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA14: 2009 Drought and Weight-for-Age, Binary DID, Urban Children

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.282**	-0.370***	-0.446***	-0.213	-0.201
	(0.114)	(0.108)	(0.135)	(0.254)	(0.175)
Drought × post × 2002 remittances					-0.486*
					(0.261)
2002 remittances × post					0.035
					(0.145)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	654	654	427	137	365

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for urban children (2005 household locality), aged 10 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include rural. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA15: 2009 Drought and BMI-for-Age, Binary DID, Urban Children

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.189**	-0.259***	-0.382***	-0.192	-0.316**
	(0.078)	(0.080)	(0.096)	(0.141)	(0.128)
Drought × post × 2002 remittances					0.043
					(0.327)
2002 remittances × post					0.049
					(0.107)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,377	1,377	992	441	882

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for urban children (2005 household locality), aged 15 or younger in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Columns, variables and controls are as in the main binary DID tables. The controls do not include rural. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA16: Continuous DID by Subgroup, Height-for-Age

	Change in the height-for-age z-score (HAZ)			
	Boys (1)	Girls (2)	Rural (3)	Urban (4)
Any 2009 deficit	-0.050	-0.108	0.101	-0.180
	(0.114)	(0.116)	(0.181)	(0.101)
	[0.672]	[0.376]	[0.614]	[0.104]
Dry vs near normal ($z \leq -1$)	-0.073	-0.009	0.033	-0.095
	(0.114)	(0.115)	(0.180)	(0.106)
	[0.533]	[0.937]	[0.876]	[0.406]
Severely dry or worse vs near normal ($z \leq -1.5$)	-0.158	-0.077	0.181	-0.241*
	(0.158)	(0.106)	(0.191)	(0.115)
	[0.344]	[0.488]	[0.519]	[0.065]
Dry × 2002 remittances	-0.364	0.089	-0.197	-0.098
	(0.226)	(0.168)	(0.211)	(0.175)
	[0.121]	[0.610]	[0.375]	[0.586]
Children	1,208	1,228	1,020	1,416
Municipalities with no 2009 deficit	42	41	27	24

Notes: This table repeats the continuous DID and discrete-dose results for the WHO height-for-age z-score (HAZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought, using sex reported in the 2005 household roster, and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). Rows, samples and definitions are as in the main continuous DID and discrete-dose tables. Children are aged 15 or younger in both survey waves. The any deficit row uses the contdid package, and the dose-group rows use the discrete-dose regression in equation 4.1 (Callaway, Goodman-Bacon, and Sant'Anna 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon, Nielsen, and Webb 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA17: Continuous DID by Subgroup, Weight-for-Age

	Change in the weight-for-age z-score (WAZ)			
	Boys (1)	Girls (2)	Rural (3)	Urban (4)
Any 2009 deficit	0.139	-0.089	0.066	0.047
	(0.143)	(0.112)	(0.202)	(0.094)
	[0.348]	[0.436]	[0.756]	[0.622]
Dry vs near normal ($z \leq -1$)	-0.221	-0.179	-0.143	-0.218**
	(0.180)	(0.111)	(0.227)	(0.103)
	[0.262]	[0.122]	[0.639]	[0.044]
Severely dry or worse vs near normal ($z \leq -1.5$)	-0.152	-0.153	0.013	-0.237
	(0.198)	(0.170)	(0.124)	(0.154)
	[0.471]	[0.428]	[0.924]	[0.168]
Dry × 2002 remittances	-0.008	-0.456	-0.144	-0.327
	(0.301)	(0.274)	(0.348)	(0.250)
	[0.977]	[0.102]	[0.705]	[0.209]
Children	556	528	424	660
Municipalities with no 2009 deficit	36	32	23	23

Notes: This table repeats the continuous DID and discrete-dose results for the WHO weight-for-age z-score (WAZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought, using sex reported in the 2005 household roster, and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). Rows, samples and definitions are as in the main continuous DID and discrete-dose tables. Children are aged 10 or younger in both survey waves. The any deficit row uses the contdid package, and the dose-group rows use the discrete-dose regression in equation 4.1 (Callaway, Goodman-Bacon, and Sant'Anna 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon, Nielsen, and Webb 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA18: Continuous DID by Subgroup, BMI-for-Age

	Change in the BMI-for-age z-score (BMIZ)			
	Boys (1)	Girls (2)	Rural (3)	Urban (4)
Any 2009 deficit	-0.008	-0.082	-0.035	-0.054
	(0.093)	(0.076)	(0.090)	(0.095)
	[0.933]	[0.294]	[0.699]	[0.593]
Dry vs near normal ($z \leq -1$)	-0.202*	-0.124	-0.147	-0.174
	(0.101)	(0.080)	(0.094)	(0.111)
	[0.054]	[0.138]	[0.155]	[0.141]
Severely dry or worse vs near normal ($z \leq -1.5$)	-0.064	-0.106	-0.129	-0.077
	(0.105)	(0.102)	(0.090)	(0.085)
	[0.548]	[0.350]	[0.309]	[0.379]
Dry × 2002 remittances	0.506	-0.301	-0.030	0.216
	(0.324)	(0.180)	(0.263)	(0.304)
	[0.142]	[0.119]	[0.915]	[0.534]
Children	1,177	1,206	1,002	1,381
Municipalities with no 2009 deficit	42	41	27	24

Notes: This table repeats the continuous DID and discrete-dose results for the WHO BMI-for-age z-score (BMIZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought, using sex reported in the 2005 household roster, and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI 2024). Rows, samples and definitions are as in the main continuous DID and discrete-dose tables. Children are aged 15 or younger in both survey waves. The any deficit row uses the contdid package, and the dose-group rows use the discrete-dose regression in equation 4.1 (Callaway, Goodman-Bacon, and Sant'Anna 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon, Nielsen, and Webb 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA19: Triple Difference by Sex, Binary DID

	HAZ (1)	WAZ (2)	BMIZ (3)
Drought × post	0.104	-0.241	-0.306***
	(0.135)	(0.175)	(0.110)
Drought × post × male	-0.163	-0.107	0.004
	(0.146)	(0.236)	(0.130)
Male × post	0.152*	0.059	-0.002
	(0.081)	(0.123)	(0.076)
P-val: pre-trend 2002–2005	0.724	0.114	0.712
Children	1,755	700	1,714

Notes: This table tests whether the effect of the 2009 drought differs for boys with a triple difference on all children, using the binary DID. Male equals 1 for boys, using sex reported in the 2005 household roster. Children are aged 15 or younger for HAZ and BMIZ and 10 or younger for WAZ in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. The regressions keep the controls of Col. 3 of the main binary DID tables and add drought × post × male, the difference in the drought effect for boys. The 2005 child and mother controls do not include child sex, which enters as male × post. The pre-trend p-value tests the 2002 lead of the triple difference in the event study on the same children, with the controls interacted with the 2002 and 2009 waves. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA20: Triple Difference by Sex, Discrete Dose

	HAZ (1)	WAZ (2)	BMIZ (3)
Dry ($z \leq -1$)	-0.009	-0.179	-0.124
	(0.115)	(0.111)	(0.080)
	[0.942]	[0.128]	[0.138]
Dry × male	-0.065	-0.041	-0.078
	(0.112)	(0.194)	(0.102)
	[0.576]	[0.834]	[0.451]
Male	0.099	0.109	0.065
	(0.064)	(0.119)	(0.064)
	[0.131]	[0.370]	[0.316]
Children	2,351	1,045	2,300

Notes: This table tests whether the effect of the 2009 drought differs for boys with a triple difference on all children, using the discrete dose. Male equals 1 for boys, using sex reported in the 2005 household roster. Children are aged 15 or younger for HAZ and BMIZ and 10 or younger for WAZ in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Regressions are on the 2005 to 2009 change in the WHO z-score, without children in wet municipalities (z of 1 or above). The omitted group is near-normal rainfall ($-1 < z < 1$). There are no controls or fixed effects. Dry × male is the difference in the effect of dry rainfall for boys. Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon, Nielsen, and Webb 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA21: Triple Difference by Rural Residence, Binary DID

	HAZ (1)	WAZ (2)	BMIZ (3)
Drought × post	0.011	-0.334***	-0.296**
	(0.127)	(0.123)	(0.117)
Drought × post × rural	0.043	0.112	-0.028
	(0.192)	(0.240)	(0.153)
Rural × post	-0.100	-0.046	0.056
	(0.103)	(0.159)	(0.096)
P-val: pre-trend 2002–2005	0.700	0.286	0.572
Children	1,755	700	1,714

Notes: This table tests whether the effect of the 2009 drought differs for rural children with a triple difference on all children, using the binary DID. Rural equals 1 if the child's 2005 household locality had fewer than 2,500 inhabitants (INEGI 2024). Children are aged 15 or younger for HAZ and BMIZ and 10 or younger for WAZ in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. The regressions keep the controls of Col. 3 of the main binary DID tables and add drought × post × rural, the difference in the drought effect for rural children. The 2002 household controls do not include rural, which enters as 2005 rural × post. The pre-trend p-value tests the 2002 lead of the triple difference in the event study on the same children, with the controls interacted with the 2002 and 2009 waves. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table OA22: Triple Difference by Rural Residence, Discrete Dose

	HAZ (1)	WAZ (2)	BMIZ (3)
Dry ($z \leq -1$)	-0.095	-0.218**	-0.174
	(0.106)	(0.103)	(0.110)
	[0.393]	[0.049]	[0.141]
Dry × rural	0.128	0.075	0.027
	(0.206)	(0.248)	(0.143)
	[0.575]	[0.796]	[0.865]
Rural	-0.141	-0.213*	-0.014
	(0.110)	(0.114)	(0.075)
	[0.220]	[0.073]	[0.847]
Children	2,351	1,045	2,300

Notes: This table tests whether the effect of the 2009 drought differs for rural children with a triple difference on all children, using the discrete dose. Rural equals 1 if the child's 2005 household locality had fewer than 2,500 inhabitants (INEGI 2024). Children are aged 15 or younger for HAZ and BMIZ and 10 or younger for WAZ in both survey waves, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009, for the same reason as in Table 1 of the paper. Regressions are on the 2005 to 2009 change in the WHO z-score, without children in wet municipalities (z of 1 or above). The omitted group is near-normal rainfall ($-1 < z < 1$). There are no controls or fixed effects. Dry × rural is the difference in the effect of dry rainfall for rural children. Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon, Nielsen, and Webb 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.